Mangamuri Venkata Pranathi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.60
Narayana Junior College	Class XII (Board of Intermediate Education)	2023	96.50%
Red Cherries School	Class X (SSC)	2021	100.00%

WORK EXPERIENCE

Disha Portal (IIT Hyderabad) | Full Stack developement :

- Contributed to the development of the Admissions Portal by designing and implementing frontend pages, integrating backend APIs, handling data flow, and ensuring smooth interaction between the UI and backend services.
- Independently designed and developed multiple modules/pages for the DISHA Portal, including frontend components, API integration, form handling, data validation, and backend connectivity.
- Worked on debugging, testing, and refining application features to improve functionality, usability, and overall application performance.

PROJECTS

Notification Framework | Node.js, Redis, BullMQ, Prisma :

- Built a centralized notification framework to send email, SMS, WhatsApp, and push notifications through a unified API.
- Developed a reliable background processing system with queued notifications, retries, rate limiting, and failure handling to ensure dependable message delivery.
- Added provider fallback and dead-letter queue (DLQ) processing to recover failed notifications and handle temporary or permanent delivery errors.
- Integrated API key management, media/attachment support, notification tracking, and a web-based interface for sending notifications and managing preferences.
- Implemented user notification preferences, allowing users to configure preferred channels, event types, and Do-Not-Disturb schedules.

ML-Based Agricultural Productivity Classification | Python, Scikit-learn, XGBoost, Pandas :

- Developed a machine learning classifier to predict agricultural field efficiency (low, medium, high) using a dataset of 112K+ training samples and 15K+ test samples.
- Performed end-to-end data preprocessing and feature engineering, including handling missing values, encoding categorical variables, and scaling numerical features.
- Implemented and evaluated multiple supervised learning models such as Logistic Regression, Random Forest, and Gradient Boosting (XGBoost/LightGBM) with k-fold cross-validation and hyperparameter tuning.
- Improved model performance through feature selection and parameter tuning, achieving better macro F1-score and balanced class predictions.

Binary Image Classification using ResNet | Python, PyTorch :

- Developed a custom ResNet-style CNN from scratch to classify synthetic geometric-object images into binary classes, adhering to the hackathon constraint of no pre-trained models.
- Designed a deep architecture with residual connections, Batch Normalization, Global Average Pooling, and progressive channels (64 -> 128 -> 256 -> 512), trained using Adam, OneCycleLR, and Kaiming initialization.
- Conducted extensive experiments with Mixup, CutMix, AdamW, SE-ResNet, CoordConv, higher-resolution inputs, ensemble methods, and object-level detection approaches to analyze model performance.
- Achieved a ~0.754 Kaggle leaderboard score, identifying that the remaining errors were primarily associated with relational reasoning between multiple objects rather than basic visual feature extraction.

Intent-Based Networking (IBN) | P4, SDN, Mininet, Python :

- Designed and implemented an Intent-Based Networking (IBN) system using P4, SDN, and Mininet.

- Enabled dynamic policy enforcement by integrating an SDN controller that computes forwarding paths using Dijkstra's algorithm and installs rules on programmable switches via P4Runtime.
- Integrated a RASA-based NLU pipeline for intent classification and entity extraction.

SKILLS

- **Programming Languages:** C++, C, JavaScript, HTML,Python.
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Web Technologies & Frameworks:** HTML, CSS, ReactJS, PHP, React, FastAPI, Node.js, Socket.io.
- **Tools & Operating Systems:** Git, GitHub, Vim, Doxygen, LaTeX,Solid Edge,Ubuntu(Linux).

RELEVANT COURSES

- **Mathematics & Other:** Calculus, Linear Algebra, Probability and Random Variables, Transform Techniques, Introduction to Statistics, Introduction to Artificial Intelligence, Digital Fabrication, Communication Skills

- **Computer Science:** Introduction to Programming, Discrete Mathematics, Software Development Fundamentals, Operating Systems,Computer Architecture, Database Management Systems, Data Structures, Algorithms, Theory of Computation, Compilers (I & II),Computer Networks , Fraud Analytics, FOML, Software Engineering,Deep Learning .

POSITIONS OF RESPONSIBILITY

Diesta Events Head | 2025 - 26 :

- Led the planning and execution of events for Diesta, coordinating logistics, teams, and event operations.
- Collaborated with students and organizing teams to drive participation and deliver successful events.

Hostel Representative | 2025-26 :

- Coordinated student concerns, communications, and hostel activities.
- Represented residents in discussions and helped facilitate effective communication and issue resolution.

Inter-IIT Coordinator | 2025:

- Worked as a coordinator in operations domain for inter-iit sports meet and staff meet 2025.

Elan and nVision Security Coordinator :

- Worked as a security coordinator in elan and nvision fest.
- Organized and managed the outsiders coming for fest and managed security for the guests came for proshows.